UNIVERSITY OF WOLLONGONG AUSTRALIA

Tutorial 09 — Optional

Root locus, gain, uncertainty and actuator limits

Optional self-study after Lecture 8. This tutorial is outside the eight-class core route and is separate from Project 2.
Suggested 60-minute tutorial: Q1 (6 min), Q2 including MATLAB (30 min), Q3 (12 min), and discussion and feedback (12 min). Extensions are additional practice. Start independently, compare reasoning with a partner, then attempt Q3 independently.

Question 1 Quick checks on root-locus meaning and feasible commands

Give a brief reason or calculation for each part.

a) Root-locus endpoints. For unity negative feedback with $L(s)=K(s+1)/[(s+2)(s+4)]$, K ≥ 0, where do the two branches start and where do they end as K tends to infinity?

b) Close a simple loop. For $L(s)=K/(s+1)$, find the closed-loop pole at K = 1 and K = 4. What happens to this pole as K increases?

c) Read a pole pair. A system has poles −3 ± 4j s^{-1}. State the exponential decay rate and damped oscillation frequency. Does the free oscillation decay?

d) Actuator clipping. A stage obeys $\ddot{x}+3\dot{x}=u$, with $u=25(r-x)-3\dot{x}$. From rest apply r = 0.020 m, but cap positive u at 0.3 m/s². Find requested command, delivered command and actual initial acceleration.

Question 2 Design a velocity-feedback inspection stage

Allow about 30 min, including the MATLAB comparison.

A calibrated inspection stage obeys $x'' + 3x' = u$, with x in m and acceleration-equivalent command u in m/s². Compare P control $u = K_p(r - x)$ with velocity feedback $u = K_p(r - x) - K_v x'$.

Target poles are $-3 \pm j4$ s⁻¹. Apply a 0.020 m reference step from rest. Require overshoot ≤ 10%, estimated 2% settling time ≤ 1.5 s and initial command ≤ 0.6 m/s².

a) Match the target polynomial and find K_p and K_v, including their units.

b) For P-only control with the same K_p, calculate the poles and percentage overshoot. Compare with the velocity-feedback design.

c) Check the velocity-feedback design against all three nominal requirements, using the usual 2% settling-time estimate. Find the initial command. Does this initial value prove that the actuator limit is met at every time?

MATLAB verification. After matching the poles, compare $K_p = 25$ with $K_v = 0$ and 3 for the 0.020 m step from rest. Submit one figure with position and requested command in separate panels; distinguish both controllers and mark 0.6 m/s². Report poles, overshoot, settling and initial command. The simulations are unsaturated. Base MATLAB is sufficient.

Question 3 A nominally good design with uncertain damping

Allow about 12 min. Use the model from Question 2; do not repeat its derivation.

Keep $K_p = 25$ from Question 2. The physical velocity coefficient b in $x'' + bx' = u$ can range from 2.4 to 3.6 s⁻¹. Compare $K_v = 3.0$ and a proposed revision $K_v = 3.6$ s⁻¹. Require overshoot ≤10% and estimated 2% settling time ≤1.5 s throughout this range.

a) Find the worst damping ratio and the corresponding overshoot and settling-time estimates for $K_v = 3.0$. Does the nominal design remain acceptable?

b) Check the proposed $K_v = 3.6$ at both endpoints of the b range. Does it meet both requirements throughout the range? Give a reason for selecting the worst endpoint.

c) Does changing K_v change the initial command for the 0.020 m step from rest? State one limit of this robustness check.

Extension A Higher gain does not always move the dominant pole left

Further practice beyond the 60-minute tutorial.

The separate normalized plant below is under unity negative feedback with proportional gain K > 0. Compare K = 1 and K = 4.

$$G(s)=\frac{s+1}{(s+2)(s+4)}$$

a) Derive the closed-loop characteristic polynomial and calculate both pole pairs.

b) Identify the dominant pole and its trend as K increases. Explain the role of the plant zero without needing detailed root-locus construction rules.

c) Calculate unit-step steady errors and reconcile better steady accuracy with the pole trend.

Method reference

Concept reference: Ogata, Modern Control Engineering, 5th ed., Sec. 6–2, printed pp. 270–278; Sec. 5–3, printed pp. 165–177.

Method source: Ogata, Modern Control Engineering, 5th ed., Problem A-5-4, pp. 235–236 (velocity feedback); Chapter 6 (root-locus interpretation). The uncertainty test and gain family are independently adapted. Page numbers refer to printed pages.

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